

MySQL - Sequential Query Language

DATA MODELING - It is a critical steps in defining data structure , creating data model to describe associations and constraints for reuse. Design , organizational structure . it may represent diagrams , symbols or text .

Terminology –

Entity – Entities are the objects or items in the business environment for which we store the data, ecommerce data- (table name) Customers , Orders , Products

Attribute – It provide a way to structure and organize the data within an entity. Name, DOB, Adreess, Email

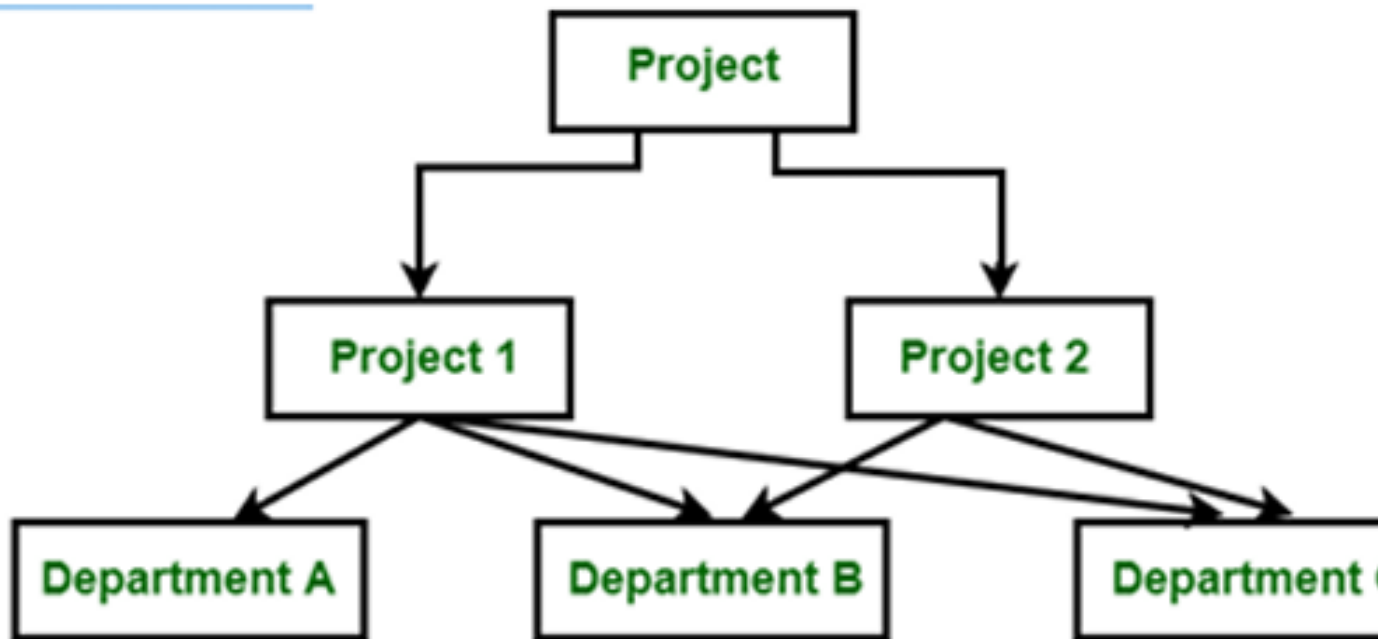
Relationship – Relationship define how entities are connected with each other

Referance Table – it is use to resove the relationship

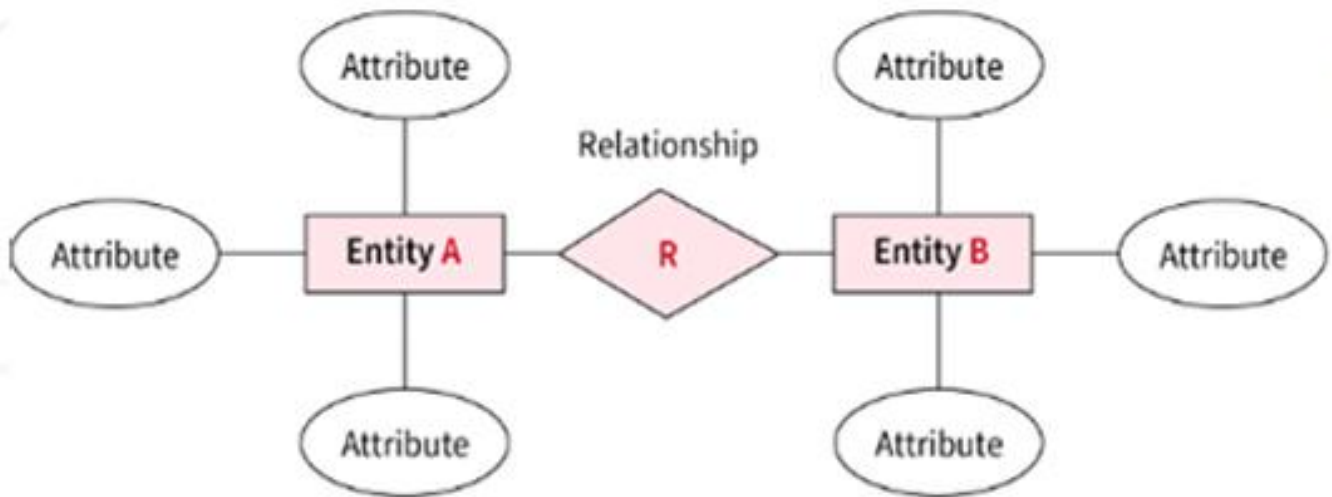
- Many to many , many to one, one to many , one to one

Schema – A schema refers to the complete description or blueprints

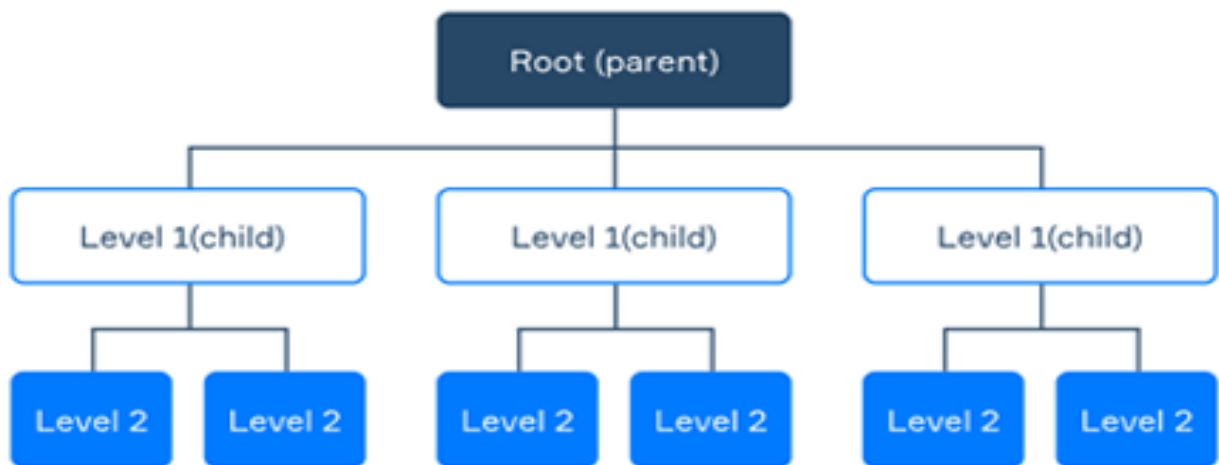
Data Model - Network Model



ER MODEL – ENTITY RELATIONSHIP MODEL



3 – Hierarchical Model



4 – Relational Model

Relation Model			
<u>OrderID</u>	<u>CustomerID</u>	<u>OrderDate</u>	<u>TotalAmount</u>
1001	1	2023-07-01	250.00
1002	2	2023-07-02	150.00
1003	1	2023-07-03	300.00
1004	4	2023-07-04	200.00

Primary Key

Tuples (Rows)

What is Database ??

A database is like a digital warehouse where data is stored, organize and managed.

-database name is not repeated or same. Once created its final .

Data – The raw information stored in a database, such as customers details, name, mo no, address.

Entities – Table

Atributes- Columns Name

Relationship –

Queries – Queries are like a search commands that allow users to extract a specific data from the database. , Search , Filter , Update, Delete, Insert ,

Different Type OF Database

1 – Relational Database - it separate into rows and columns form in the table. SQL , ORACLE , MYSQL , MICROSOFT SQL ,

2 – NoSQL Database – No relational database – (wikipedia)
- MONGODB , CASSENDRA , COUCHBASE

3 – OBJECT ORIENTED DATABASE – ObjectDb , db4o

4 – Hierarchical Database – MIS – Management Information System – hospital

5 – Time Series Database – influxDB ,

RDBMS – RELATIONAL DATABASE MANAGEMENT SYSTEM

DBMS – DATABASE MANAGEMENT SYSTEM

MYSQL – RELATIONSHIP

- **One – to – one**

You are having adhar card , from adhar office.

- **One to Many**
- **From Amazon One customer can purchase the multiple / many products**

Many to One

- **Hospital Database – One surgeon can performs many operations**

Many to Many

- **Many students can select the many subjects**

SQL – COMMANDS

SQL – CONSTRAINTS

SQL – CLAUSES

SQL – JOINTS

KEYS -

